

Ibrahim Khadraoui

Senior AI / ML Systems Engineer — Edge Inference & Embodied AI

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PROFILE

ML Systems Engineer with over eight years of experience developing and deploying AI systems across embedded platforms, robotics, and edge devices. Recent work focuses on large language models (LLMs) and Vision Language Action (VLA) models, spanning architecture integration, model optimization, quantization, compiler and runtime development, and production deployment. Proven track record of translating research into production-ready systems, achieving millisecond-scale inference while maintaining numerical parity with reference implementations.

SKILLS

Inference optimization : MLX, TensorRT, ggml/GGUF

Model porting & validation : hand-written C++/ggml transformer runtimes; upstream llama.cpp contributor

Model adaptation : LoRA fine-tuning of LLMs; fine-tuning of object detection models

LLM / VLM serving : vLLM, SGLang, llama.cpp, TensorRT-LLM; TTFT and throughput benchmarking; self-hosted ASR/TTS pipelines

Robotics & control : Mobile ALOHA robot, inverse kinematics, ROS 2 / MoveIt

Systems & infrastructure : Linux internals, Docker/Compose, systemd/udev, GStreamer/ffmpeg/WebRTC/RTSP, MQTT/ZeroMQ

Languages : Python (expert) · C++/C (strong, low-level)

EXPERIENCE

Technology Innovation Institute (TII)

Senior AI Engineer (since Nov 2025)

Mar. 2022 – Present

Abu Dhabi, UAE

Previously: AI Engineer

Focus: Edge AI & Embodied Intelligence

2024 – Present

- **Work on the inference and deployment** path for FalconVLA, TII's 8B vision-language-action model family, from PyTorch research checkpoint to running policy on physical bimanual robot hardware
- **Optimized a model's TensorRT path to run 1.72× faster than PyTorch** (101.4 ms vs 174 ms per query) at identical action accuracy
- **Authored the upstream Falcon-H1 implementation in llama.cpp** — [merged PR ggml-org/llama.cpp#14534](#), adding initial support for the Falcon-H1 hybrid-head model family to the project used worldwide for local LLM inference; also implemented Falcon3VLM support — vision tower, projector, tokenizer and HF→GGUF converters C++ · ggml · llama.cpp
- **Ported the same models to MLX** for native inference on Apple silicon Python · MLX
- **Contributed to the Falcon-H1 model series** — co-author of the [technical report](#) (arXiv:2507.22448)
- **Fine-tuned large language models with LoRA** and detection models for downstream deployment
- **Built a voice-to-action platform** for conversational robot control — LangGraph agent with confirm-before-execute and VLM scene grounding, self-hosted ASR/TTS, always-on three-camera WebRTC streaming, 3D URDF visualisation
- **Mentored new joiners and interns**, teaching engineering practice and sharing knowledge across the team — several stay in contact long after moving on
- **Reviewed code delivered by external collaborations** building a video surveillance pipeline combining a vision-language model with an object detection model — acting as technical reviewer on their pull requests

Focus: UAV, Immersive & Streaming Systems

2022 – 2024

- **Engineered a VR-controlled UAV system operated remotely over 70 km on a 5G network**, with live 360° video, built end to end: airframe and electronics, cloud-server control pipeline, and haptic feedback conveying real-time telemetry and the sensation of drone movement; **demonstrated publicly at GITEX Dubai 2022** and featured by [NAS Daily](#)
- **Designed and built a smart glove from scratch** — board layout, soldering and firmware — used as the drone's control interface and adapted to further use cases
- **Collaborated with the Live in Five team on the A2RL autonomous racing competition**: decoded the full EMM protocol and developed the camera-view switching algorithm used for the viewer experience
- Built a [VR application in A-Frame](#) for visualising Gaussian splatting models produced at TII
- **Awarded the Nova Award** for the quarter, in a research environment of senior scientists

Independent contractor

- Started while still a student, working directly with companies to build smart IoT solutions end to end — schematic capture and PCB layout, through firmware, up to the platform running on top
- Delivered complete hardware and firmware prototypes for universities, researchers and startups, covering the full path from concept to a reliable working device

SELECTED PROJECTS

ATC — real-time drone detection and tracking

- Modular backend architecture (RKNN / ONNX) running a fine-tuned YOLOv5 + ByteTrack on an Orange Pi 5 Pro NPU with INT8, C++ detector service over libav, RTSP→WebRTC delivery C++ · RKNN · MediaMTX · WebRTC · MQTT · PX4 · Gazebo · IsaacSim

Sheep Counting Benchmark — multi-paradigm counting evaluation platform

- Containerised benchmark putting **20+ model variants** behind one contract — detection, density estimation, counting specialists and VLM pointing — and ranking them on MAE / RMSE / MAPE and latency, with detection overlays in an interactive UI. Adding a model needs no change to the orchestrator or frontend Python · FastAPI · Docker

PUBLICATIONS

- **ALRM: Agentic LLM for Robotic Manipulation** — arXiv:2601.19510, January 2026
- **Falcon-H1: A Family of Hybrid-Head Language Models Redefining Efficiency and Performance** — arXiv:2507.22448, July 2025
- **360-Degree Video Super Resolution and Quality Enhancement Challenge: Methods and Results** — arXiv:2411.06738, November 2024
- **Energy Cost of Coding Omnidirectional Videos using ARM and x86 Platforms** — GMSys '24, 2nd ACM Green Multimedia Systems Workshop @ MMSys, April 2024
- **Locomotion-based UAV Control Toward the Internet of Senses** — IEEE Transactions on Circuits and Systems II: Express Briefs, vol. 70, no. 5, pp. 1804–1808, May 2023
- **Virtual Reality and Multimedia Towards the Internet of Senses** — Wireless World Research Forum, 2022
- Full list: [Google Scholar profile](#)

EDUCATION

University of Science and Technology Houari Boumediene (USTHB)

Sept. 2021

Master's Degree, Embedded Systems Electronics

- **Thesis:** heavy-carrying mobile robot with human-following capability using Ultra-Wideband localisation and ROS

University of Science and Technology Houari Boumediene (USTHB)

Sept. 2019

Bachelor's Degree, Electronics

- **Project:** smart sorting machine with RFID identification and database-backed inventory

RECOGNITION

- **Upstream contributor to llama.cpp** — authored the merged Falcon-H1 model family implementation ([PR #14534](#))
- **Nova Award** — quarterly research excellence award, TII
- **GITEX Dubai 2022** — public demonstration of 70 km VR-controlled UAV with 360° video streaming over 5G
- **Featured by NAS Daily** — 5G remote drone control system

LANGUAGES

Arabic (native) · French (fluent) · English (professional)